

Basic Electrical Engineering

MODULE 4 — PART 1: COMPLETE NOTES

Module 4: AC Machines — Transformer and Three-Phase Induction Motor

For First-Year Students — written in simple, easy language
Course: Basic Electrical Engineering · Course Code: EE100

Module 4 — what it covers

- Transformer: construction, working principle, EMF equation and transformation ratio
- Transformer: types, losses, efficiency and applications
- Three-phase induction motor: construction (squirrel-cage and wound rotor)
- Rotating magnetic field, synchronous speed, principle of operation and slip
- Torque characteristics and applications

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1. Introduction to the Transformer

A transformer is a **static machine** used for transforming power from one circuit to another **without changing the frequency**. It can raise or lower the voltage or current in an AC circuit and can isolate circuits from each other. Because it has no rotating or moving part, it is a static device. It operates only on an AC supply and works on the principle of **mutual induction**.

- It enables electrical energy to be transmitted over great distances economically.
- It steps up the voltage for transmission and steps it down for use.
- It cannot change the frequency of the supply.



Figure 1: A single-phase transformer.

1.1 Mutual Induction

The alternating current through a winding produces a continually changing (alternating) flux that surrounds the winding. If another winding is brought close to it, some of this alternating flux links the second winding. As the flux is continually changing in magnitude and direction, there is a changing flux linkage in the second winding, and by Faraday's law of electromagnetic induction an emf is induced in it. If the circuit of this second winding is closed, a current flows through it.

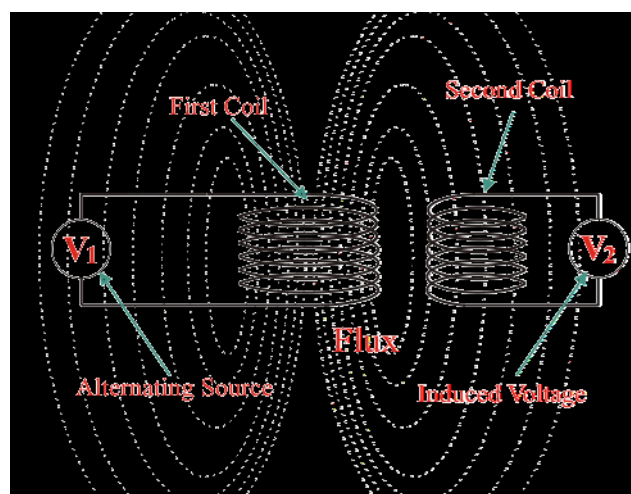


Figure 2: Mutual induction between two coils.

2. Construction of a Transformer

A transformer has two windings wrapped around a laminated core:

- **Primary winding** — the winding which receives electrical power from the source.
- **Secondary winding** — the winding which gives the desired output voltage by mutual induction.

The windings are made of insulated copper wire or strip and are insulated from each other and from the core by press-board or Bakelite cylinders. The **core** provides a low-reluctance path so that the maximum flux produced by the primary is linked with the secondary. The core is laminated to reduce eddy-current loss.

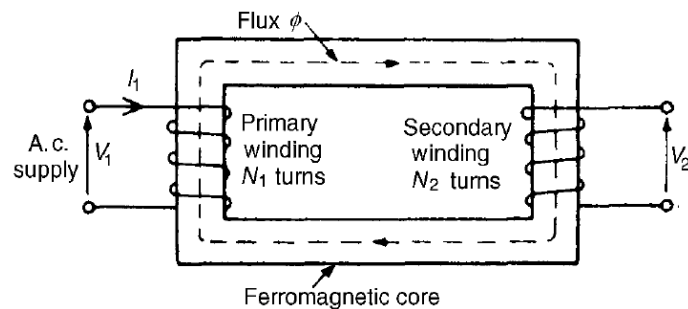


Figure 3: Construction of a transformer.

Single-phase transformer — basic arrangement

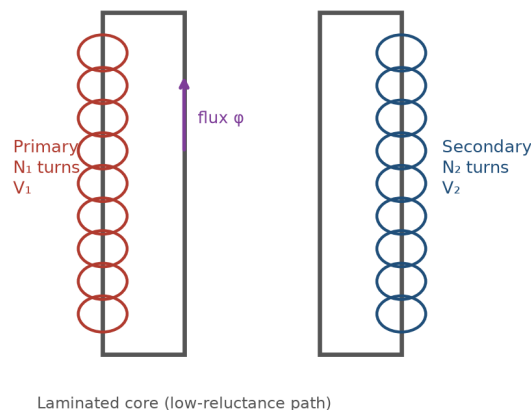


Figure 4: Basic arrangement of a single-phase transformer.

2.1 Core-type and Shell-type Construction

In a **core-type** transformer the windings are wrapped around the two sides (limbs) of a laminated square core, and in a **shell-type** transformer the windings are wrapped around the centre leg of the core. The core-type is generally used for high-voltage transformers and the shell-type for low-voltage transformers.

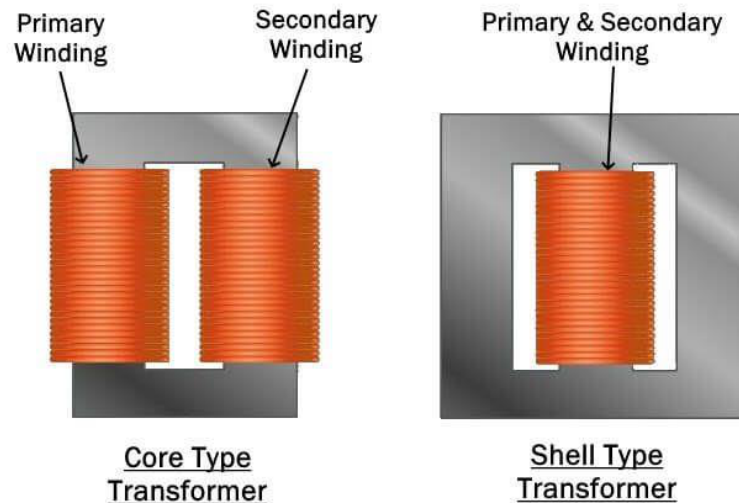


Figure 5: Core-type and shell-type construction.

3. Principle of Operation

When the secondary is on open-circuit and an alternating voltage V_1 is applied to the primary winding, a small current called the **no-load current I_0** flows, which sets up a magnetic flux in the core. This alternating flux links both the primary and the secondary windings and induces emfs E_1 and E_2 in them.

In an ideal transformer the rate of change of flux is the same for both windings, so the induced emf per turn is constant. Assuming no losses, $E_1 = V_1$ and $E_2 = V_2$, hence:

$$V_2 / V_1 = E_2 / E_1 = N_2 / N_1 = K$$

V_2/V_1 is called the **voltage ratio** and N_2/N_1 the **turns ratio** or **transformation ratio K** . If $N_2 < N_1$ then $V_2 < V_1$ and the transformer is a **step-down** transformer; if $N_2 > N_1$ then $V_2 > V_1$ and it is a **step-up** transformer.

When a load is connected across the secondary, a current I_2 flows. For an ideal transformer (100% efficient) the input power equals the output power:

$$V_1 I_1 = V_2 I_2 \quad \text{so} \quad V_2/V_1 = I_1/I_2 = N_2/N_1 = K$$

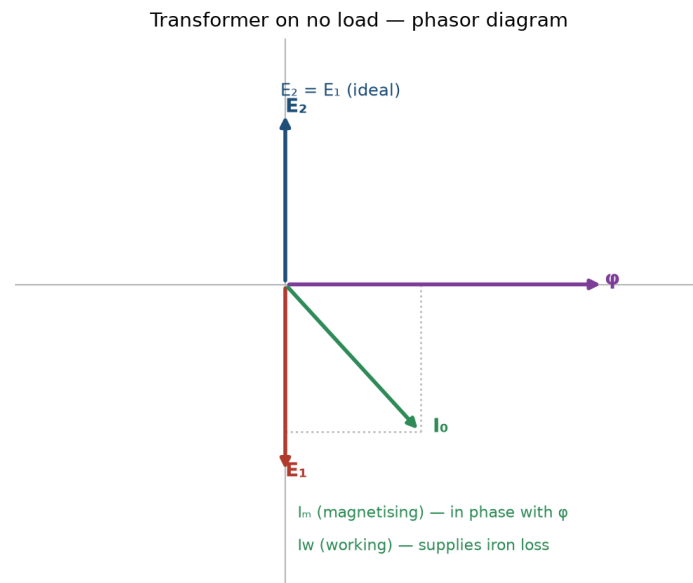


Figure 6: Phasor diagram of a transformer on no load.

4. EMF Equation of a Transformer

Let the maximum flux in the core be ϕ_m weber and the supply frequency be f hertz. The average rate of change of flux is $4f\phi_m$ and the rms value is 1.11 times this, so the emf induced in a winding of N turns is:

$$E = 4.44 f N \phi_m \text{ volts}$$

where $\phi_m = B_m \times A$ (maximum flux density $\times$ net core area). For the primary and secondary:

$$E_1 = 4.44 f N_1 \phi_m \quad E_2 = 4.44 f N_2 \phi_m$$

Dividing, $E_2/E_1 = N_2/N_1$, so the emf per turn is the same in both windings.

$$\begin{aligned} \text{Hence rms value} &= \text{form factor} * \text{average value} \\ &= 1.11 * \text{average value} \end{aligned}$$

$$\begin{aligned} \text{Thus rms e.m.f. induced in each turn} &= 1.11 * 4f\phi_m \text{ volts} \\ &= 4.44f\phi_m \text{ volts} \end{aligned}$$

Therefore, rms value of e.m.f. induced in primary,

$$E_1 = 4.44f\phi_m N_1 \text{ volts}$$

and rms value of e.m.f. induced in secondary,

$$E_2 = 4.44f\phi_m N_2 \text{ volts} \quad E_2 / N_2 = E_1 / N_1$$

Figure 7: EMF equation of a transformer.

Example: for $f = 50$ Hz, $N = 400$ turns and $\phi_m = 0.02$ Wb, $E = 4.44 \times 50 \times 400 \times 0.02 = 1776$ V.

5. Types of Transformer

Classified by	Types
Construction	Core type (windings on the two limbs) and shell type (windings on the centre leg).
Voltage level	Step-up transformer (raises voltage) and step-down transformer (lowers voltage).
Use	Power transformer (transmission) and distribution transformer (local supply).

Whether the transformer increases or decreases the voltage depends only on the relative number of turns on the primary and secondary sides.

6. Losses in a Transformer

In any electrical machine, "loss" is the difference between input power and output power. Since a transformer is a static device, mechanical losses (windage and friction) are absent; it has only **electrical losses** — iron (core) losses and copper losses.

Loss	Cause and reduction
Copper loss	Due to the ohmic resistance of the windings: $I_1^2 R_1 + I_2^2 R_2$. It varies with the load.
Hysteresis loss	Due to reversal of magnetisation in the core; depends on the volume and grade of the iron, the frequency and the flux density. Reduced by using nickel-iron alloys whose hysteresis loops have a small area.
Eddy-current loss	Emf induced in the core itself sets up circulating eddy currents, causing loss and heating. Reduced by laminating the core or increasing the resistivity of the core material.

Hysteresis loss and eddy-current loss depend on the magnetic properties of the core material, so they are together called **core losses** or **iron losses**.

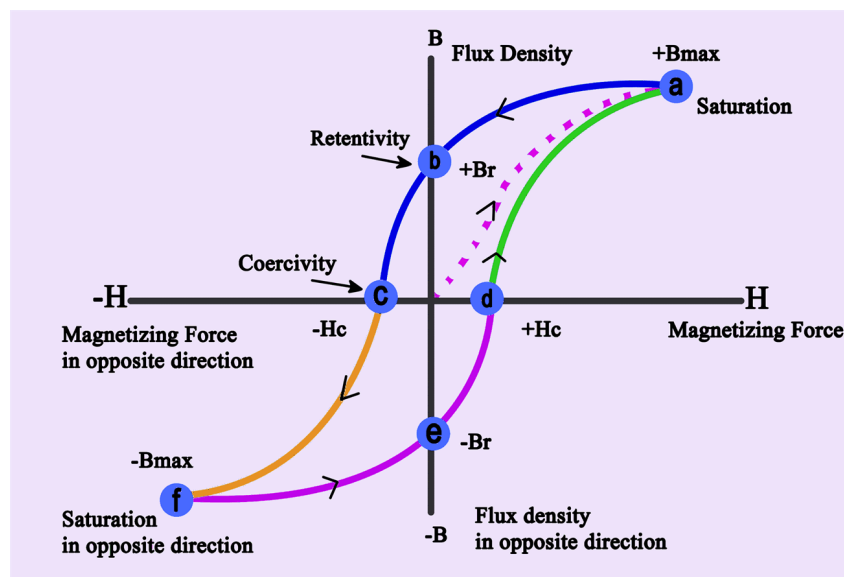


Figure 8: Iron (core) losses and copper loss.

7. Efficiency of a Transformer

The efficiency of a transformer is the ratio of output power to input power:

$$\eta = \text{Output} / \text{Input} = (\text{Input} - \text{Losses}) / \text{Input}$$

Transformers are among the most efficient electrical devices; most have a full-load efficiency of 95% to 98.5%. Because the output and input are nearly equal, it is impractical to find the efficiency from output/input, so the losses method is used instead.

Example: for an output of 10 kW and total losses of 500 W, $\eta = 10000/(10000 + 500) = 95.24\%$.

8. Three-Phase Induction Motor — Introduction

Three-phase induction motors are the most common and frequently encountered machines in industry. Their main features are:

- Simple design, rugged construction, low price and easy maintenance.
- A wide range of power ratings — from fractional horsepower to about 10 MW.
- They run essentially at constant speed from no-load to full load.
- Their speed depends on the frequency of the power source, so variable speed control needs a variable-frequency drive.

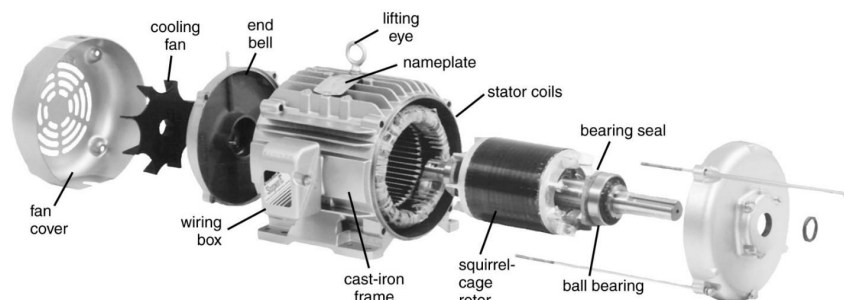


Figure 9: A three-phase induction motor.

9. Construction of a Three-Phase Induction Motor

An induction motor has two main parts:

- **Stator** — a stationary steel frame supporting a hollow cylindrical core made of stacked laminations with evenly spaced slots that hold the stator winding.
- **Rotor** — a revolving part made of punched laminations stacked to form rotor slots that hold the rotor winding.

Depending on the rotor design there are two basic types:

Rotor type	Description
Squirrel-cage rotor	Conducting bars (copper or aluminium) laid in the slots and shorted at both ends by shorting (end) rings, forming a cage-like circuit.
Wound (slip-ring) rotor	A complete set of three-phase windings exactly like the stator, usually Y-connected, with the three ends brought out to three slip rings on the shaft so the rotor circuit is accessible.

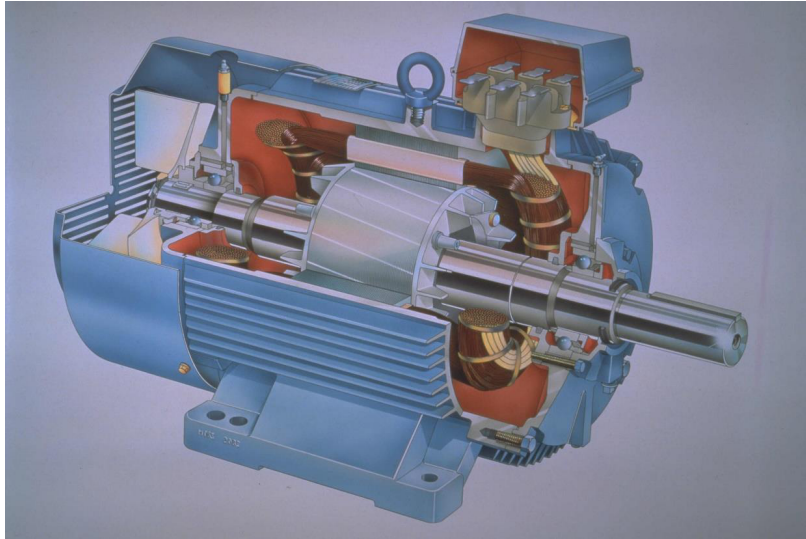


Figure 10: Squirrel-cage rotor.

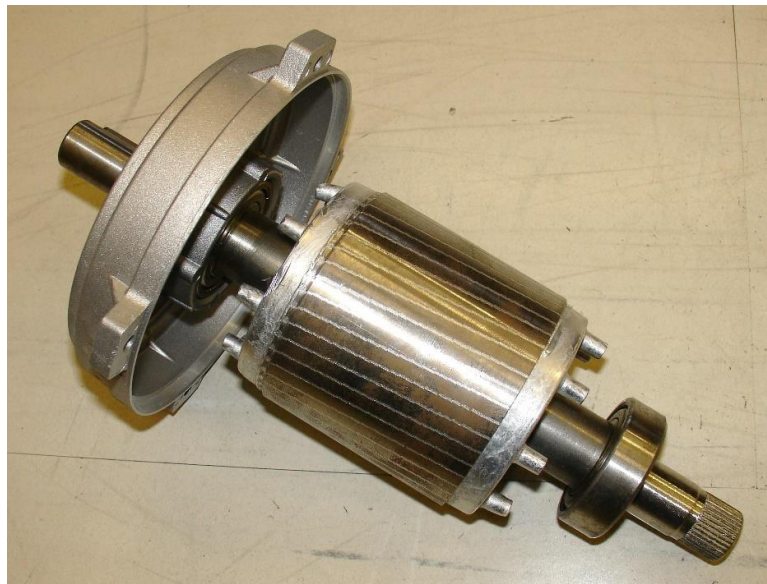


Figure 11: Wound (slip-ring) rotor.

10. Rotating Magnetic Field and Synchronous Speed

Balanced three-phase windings, mechanically displaced by 120° from each other and fed by a balanced three-phase supply, produce a **rotating magnetic field of constant magnitude**. This field rotates at a speed called the **synchronous speed**:

$$n_s = 120 f / P \text{ rpm}$$

where f is the supply frequency and P is the number of poles.

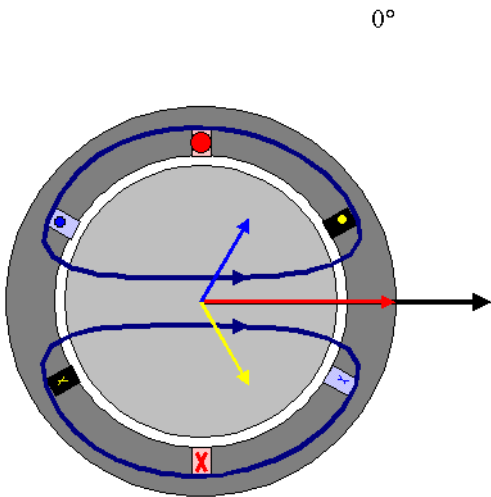


Figure 12: The rotating magnetic field.

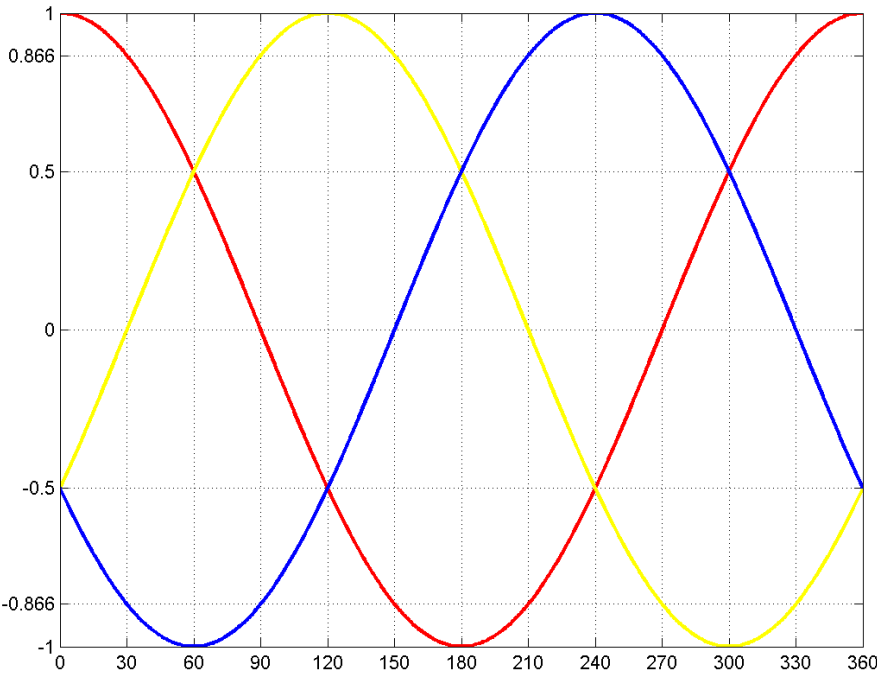


Figure 13: Interaction of the rotating field with the rotor.

11. Principle of Operation

Consider a simple rotor with one short-circuited coil placed inside the stator. Initially the rotor is stationary. The moment the stator supply is switched on, currents begin to flow and the rotating magnetic field is established. The relative motion between the moving field and the stationary rotor conductors induces an emf in them (Faraday's law). Since the conductors are short-circuited by the end rings, currents start flowing in them.

These rotor currents create their own magnetic fields, which interact with the rotating stator field to produce forces on the individual conductors and a net **rotor torque**. The rotor starts to accelerate, which lowers the relative speed between the field and the rotor conductors, so the induced emfs, currents and torque fall. The rotor continues to accelerate until the electrical torque exactly equals the mechanical load torque on the shaft.

At this point the rotor runs at a speed slightly slower than the rotating field. This small difference in speed is necessary: a torque can exist only if currents flow in the rotor conductors, and those currents need an induced emf, which in turn needs a difference between the rotor speed and the speed of the rotating field.

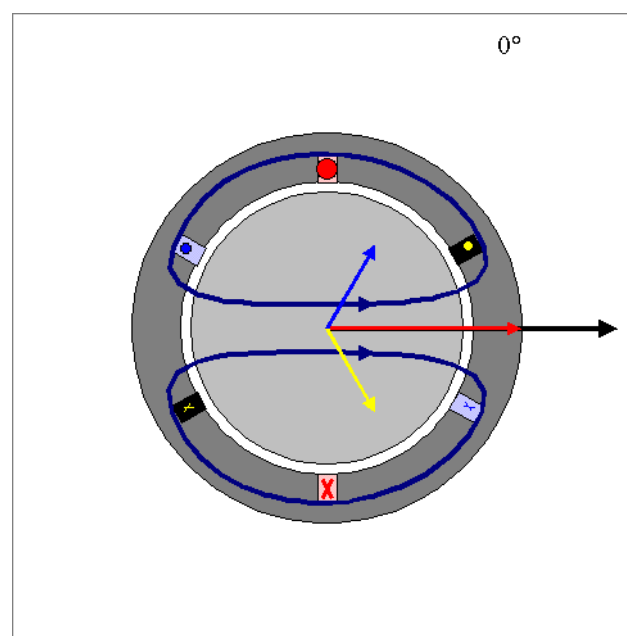


Figure 14: Rotor conductors cutting the rotating field.

12. Slip

The difference between the synchronous speed and the actual rotor speed is expressed as a ratio called the **slip**:

$$s = (n_s - n) / n_s$$

where n_s is the synchronous speed and n the actual rotor speed. The rotor speed is therefore $n = (1 - s) n_s$. For most machines the slip is about 0.01 (1%) at no load and about 0.10 (10%) at full load. The rotor frequency is $f_r = s f$.

$$n_{sync} = \frac{120 f_e}{P} \text{ rpm}$$

Figure 15: Synchronous speed of the rotating field.

13. Torque and Applications

The torque developed by an induction motor is proportional to the product of the flux and the rotor current in phase with it: $T \propto \phi I_2 \cos \phi_2$. The torque rises with slip up to a maximum value (the breakdown torque) and then falls, as shown by the torque-slip characteristic.

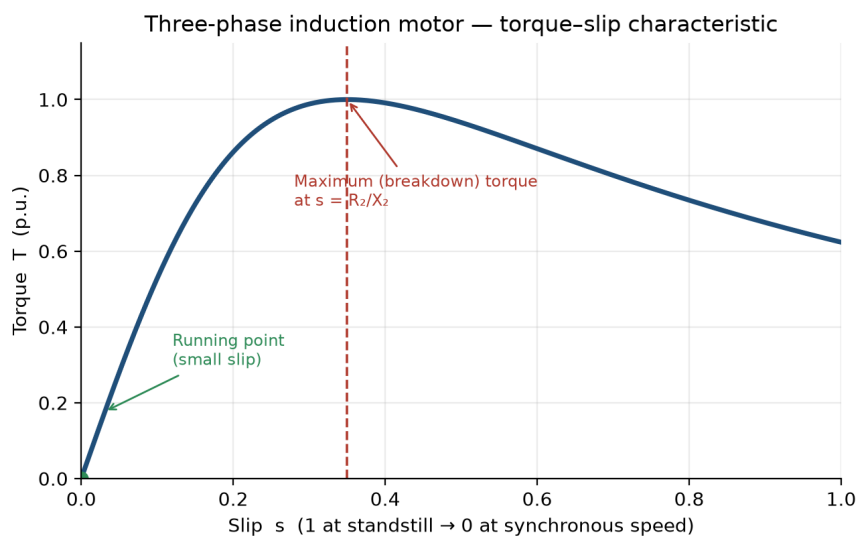


Figure 16: Torque-slip characteristic of an induction motor.

Applications

Three-phase induction motors are used for lifts, cranes, hoists, large-capacity exhaust fans, lathe machines, crushers, oil-extracting mills, textile mills and many other industrial drives.

14. Solved Numerical Examples

Example 1: Turns ratio and secondary voltage

Problem: a transformer has 500 primary turns and 100 secondary turns, with 230 V applied to the primary. Find the secondary voltage and state the type.

$$K = N_2/N_1 = 100/500 = 0.2; \quad V_2 = K V_1 = 0.2 \times 230 = 46 \text{ V (step-down)}$$

Example 2: EMF equation

Problem: a transformer has 400 primary turns and a maximum core flux of 0.02 Wb at 50 Hz. Find the primary emf.

$$E_1 = 4.44 f N_1 \phi_m = 4.44 \times 50 \times 400 \times 0.02 = 1776 \text{ V}$$

Example 3: Turns from core area

Problem: a 10 kVA, 3300/240 V, 50 Hz transformer has a core area of 300 cm² and a flux density of 1.3 T. Find the primary and secondary turns.

$$\phi_m = B_m A = 1.3 \times 300 \times 10^{-4} = 0.039 \text{ Wb}$$

$$N_1 = E_1 / (4.44 f \phi_m) = 3300 / (4.44 \times 50 \times 0.039) = 381 \text{ turns}$$

$$N_2 = E_2 / (4.44 f \phi_m) = 240 / (4.44 \times 50 \times 0.039) = 28 \text{ turns}$$

Example 4: Efficiency

Problem: a transformer delivers 10 kW with total losses of 500 W. Find its efficiency.

$$\eta = \text{Output} / (\text{Output} + \text{Losses}) = 10000 / 10500 = 95.24\%$$

Example 5: Synchronous speed and slip

Problem: a 4-pole, 50 Hz induction motor runs at 1440 rpm. Find its synchronous speed and slip.

$$n_s = 120 f / P = 120 \times 50 / 4 = 1500 \text{ rpm}; \quad s = (1500 - 1440) / 1500 = 0.04 = 4\%$$

15. Quick Reference — Key Formulae and Definitions

Quantity	Formula	Notes
Transformation ratio	$K = N_2/N_1 = E_2/E_1 = V_2/V_1 = I_1/I_2$	Turns ratio
EMF equation	$E = 4.44 f N \phi_m$	$\phi_m = B_m A$
Ideal transformer	$V_1 I_1 = V_2 I_2$	100% efficient
Copper loss	$I_1^2 R_1 + I_2^2 R_2$	Varies with load
Efficiency	$\eta = \text{Output} / \text{Input} = (\text{Input} - \text{Losses}) / \text{Input}$	95–98.5% typical
Synchronous speed	$n_s = 120 f / P$	rpm
Slip	$s = (n_s - n) / n_s$	0.01 no-load, 0.10 full load
Rotor speed	$n = (1 - s) n_s$	slightly below n_s
Rotor frequency	$f_r = s f$	Hz

Key Definitions at a Glance

Term	Definition
Transformer	A static device that transfers electrical power from one circuit to another without change of frequency, using mutual induction.
Primary / secondary winding	The winding that receives power / the winding that delivers the output.
Transformation ratio	The ratio of secondary to primary turns (and of emf, voltage).
Step-up / step-down	A transformer that raises / lowers the voltage.
Iron (core) loss	The hysteresis and eddy-current losses in the core; constant with load.
Copper loss	The I^2R loss in the windings; varies with the load.
Induction motor	An AC motor in which the rotor current is produced by induction from the rotating stator field (no direct connection).
Squirrel-cage rotor	A rotor of conducting bars shorted at both ends by end rings.
Slip-ring rotor	A rotor with a three-phase winding brought out to three slip rings.
Synchronous speed	The speed of the rotating magnetic field, $n_s = 120f/P$.
Slip	The fractional difference between synchronous speed and rotor speed.